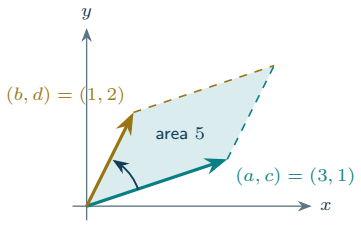
LINEAR ALGEBRA ON ONE PAGE

The determinant in its many guises

One number that is an oriented area, a difference of two squares, a product of eigenvalues and a count of tilings.

1 / THE RULE AND THE PICTURE

The determinant is the oriented area of the parallelogram spanned by the columns



For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the columns (a, c) and (b, d) span a parallelogram whose signed area is

$$\boxed{\det A = ad - bc}$$

Here $3 \cdot 2 - 1 \cdot 1 = 5$. The sign records orientation, positive when the turn from the first column to the second is anticlockwise, so swapping the columns gives -5, and a determinant of 0 means the columns lie on one line and the parallelogram has collapsed.

2 / FAMILIES OF MATRICES / ONE FORMULA, MANY NAMES

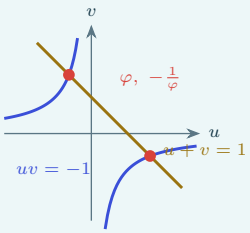
Restricting the shape of A turns $ad - bc$ into a norm

matrix	determinant	name	example
$\begin{pmatrix} a & b \\ b & a \end{pmatrix}$	$a^2 - b^2 = (a-b)(a+b)$	difference of two squares	$\begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix} \to 16 = 2 \cdot 8$
$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$	$a^2 + b^2$	Gaussian norm	$\begin{pmatrix} 3 & -2 \\ 2 & 3 \end{pmatrix} \to 13$
$\begin{pmatrix} a & 2b \\ b & a \end{pmatrix}$	$a^2 - 2b^2$	Pell norm	$\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} \to 1$
3×3 circulant (a, b, c)	$a^3 + b^3 + c^3 - 3abc$	Eisenstein norm times $a + b + c$	$(1, 2, 3) \to 18 = 6 \cdot 3$

Each 2×2 row is multiplication by $a + bj$ with $j^2 = +1,\ -1,\ +2$, so its determinant is the norm of $a + bj$.

3 / THE CHARACTERISTIC POLYNOMIAL / EIGENVALUES ON A HYPERBOLA

The eigenvalues are where $u + v = \mathrm{tr}A$ meets $uv = \det A$



Since $\det(\lambda I - A) = \lambda^2 - \mathrm{tr}(A)\,\lambda + \det A$, the determinant is the product of the two eigenvalues and the trace is their sum. For $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ they are $a \pm b$, which is why $a^2 - b^2$ factors, and for $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ they are $a \pm ib$, with product $a^2 + b^2$. The matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ has trace 1 and determinant -1, so its eigenvalues are the points in the picture, φ and $-\frac{1}{\varphi}$.

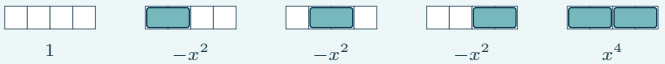
4 / TWO PROPERTIES THAT MAKE IT WORK / PRODUCTS AND SHEARS

$\det(AB) = \det A \det B$, and adding a multiple of one row to another leaves $\det$ unchanged

Multiplying rotation-scaling matrices (a, b) and (c, d) turns a matrix product into a number product, $(a^2+b^2)(c^2+d^2) = (ac-bd)^2+(ad+bc)^2$, so $5 \cdot 13 = 65 = (-4)^2 + 7^2$. Because row shears keep $\det$, elimination reads it off as the product of the pivots: for $3x + 2y = 16$ and $2x + 5y = 19$ the pivots are 3 and $\frac{11}{3}$, and the determinant is 11.

5 / A DETERMINANT THAT COUNTS / THE COMPLEX FIBONACCI GENERATOR

With 1 on the diagonal and i beside it, the determinant becomes the Fibonacci number F_{n+1}



n	1	2	3	4	5	6	7
$K_n(i)$	1	2	3	5	8	13	21

Expanding the determinant with 1 on the diagonal and x beside it gives $K_n = K_{n-1} - x^2 K_{n-2}$, and $x = i$ turns the minus into the plus of Fibonacci. Each term is a tiling of a row of squares by squares and dominoes, each domino weighing $-x^2 = +1$, so for $n = 4$ the five tilings give $1 + 3 + 1 = 5$.

6 / LINKS TO OTHER SHEETS / WHERE EACH GUISE IS DEVELOPED

in this sheet	developed in
oriented area, shears and pivots	*Linear simultaneous equations and the LCM*; *The LCM method as Gaussian elimination*
eigenvalues on a hyperbola	*Solving a quadratic as a line meeting a hyperbola*; *Cardano's cubic as a hyperbola*
Gaussian, Pell and Eisenstein norms	*Cyclotomic fields and the arcs of inscribed polygons*
tridiagonal determinant and tilings	*Why i makes Fibonacci inevitable*; *Metallic means as differences of squares*

ASIDE / FOR THE CURIOUS

The eigenvalues of the Fibonacci matrix are $1 + 2i\cos\frac{k\pi}{n+1}$, and their product is F_{n+1}.

ASIDE / CHECK IN PYTHON

`np.linalg.det` of the 4×4 matrix with 1 and i gives 5.

Convention. tr is the trace, φ the golden ratio, $F_1 = F_2 = 1$.

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